# Patricia Al-Bahar

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## EDUCATION

**University of Cambridge** | *MSc Computer Science* 2010-2012

- Designed distributed systems thesis on Byzantine fault-tolerant consensus protocols, achieving 99.97% fault tolerance with sub-50ms latency
- Led undergraduate CS mentorship program, coaching 15+ junior students in systems design and algorithms

**University of Edinburgh** | *BSc Software Engineering (Hons)* 2006-2010

- Graduated with distinction; received prize for best systems programming capstone project
- Organized annual hackathon attracting 200+ computer science students and industry professionals

## EXPERIENCE

**Stripe** | *Principal Backend Engineer* 2021-Present

- Architected and led development of payment processing platform handling $500B+ annual transaction volume with 99.99% uptime SLA, serving 2M+ businesses globally
- Reduced payment processing latency by 42% through implementation of adaptive connection pooling and optimized database query patterns, improving customer satisfaction scores by 38%
- Mentored 12 senior engineers across payments and infrastructure teams, promoting 3 to staff engineer levels within 18 months

**Palantir Technologies** | *Senior Principal Engineer, Data Platforms* 2018-2021

- Built core data pipeline framework processing 2TB+ daily across 50+ enterprise customers, reducing pipeline failure rates from 12% to <0.5% through improved orchestration
- Designed event sourcing architecture supporting 5000+ concurrent data transformations with <2 second latency, enabling real-time analytics for government agencies
- Established backend platform engineering excellence program across 60-person team, reducing deployment cycle time by 67% and increasing system reliability from 95% to 99.8%

**Cloudflare** | *Staff Backend Engineer, Infrastructure* 2015-2018

- Designed distributed DNS resolution system serving 500M+ daily queries with sub-5ms average response time, improving global latency performance by 31%
- Led 8-engineer effort to refactor rate limiting middleware, handling 3x traffic growth while reducing memory footprint by 56% and cutting infrastructure costs by $2.4M annually
- Pioneered internal observability platform using custom metrics collection, reducing mean time to resolution (MTTR) for incidents by 52% and preventing 18+ potential P1 outages

**Figma** | *Principal Software Engineer* 2019-2021

- Architected backend services for real-time collaboration engine supporting concurrent editing by 100+ users with <100ms synchronization latency across global datacenters
- Optimized database query performance for document persistence, reducing P99 query time by 73% and enabling support for 10x larger design files
- Implemented comprehensive API gateway and authentication layer, improving security compliance score from 78% to 97% and reducing unauthorized access incidents to zero

**Wise (formerly TransferWise)** | *Senior Staff Engineer, Payments* 2012-2015

- Developed settlement engine processing 40+ currency pairs and $30B+ in annual transfers with guaranteed delivery-vs-payment semantics across 190+ countries
- Refactored legacy monolith into microservices architecture, reducing deployment time from 6 hours to 12 minutes and increasing feature velocity by 4x
- Championed operational excellence initiatives including chaos engineering practices, reducing critical incidents by 64% and improving customer trust scores measurably

## PROJECTS

**AsyncRaft - Distributed Consensus Protocol** | *Rust, Protocol Buffers, gRPC, TiKV* 2022-Present

- Engineered open-source Raft consensus implementation achieving 100K+ ops/second throughput with configurable consistency guarantees, accumulated 2.8K GitHub stars
- Reduced consensus latency from 250ms to 45ms through novel leader election optimization, published findings in 3 peer-reviewed systems conferences

**EventTrace - Distributed Tracing for Edge Computing** | *Go, Jaeger, ClickHouse, Kubernetes* 2020-2022

- Built lightweight distributed tracing solution for edge infrastructure reducing observability overhead by 71% while maintaining 99.9% accuracy in span correlation

- Contributed as core maintainer with 1.2K GitHub stars; solution adopted by 8+ Fortune 500 companies processing 15B+ trace events monthly

## EXTRACURRICULAR

**DevOps & Systems Design Meetup - London** | *Co-Founder & Regular Speaker* 2017-Present

- Grew community to 2,300+ members through monthly talks on distributed systems, infrastructure patterns, and scalability challenges
- Delivered 8 keynote talks on designing fault-tolerant systems, accumulating 50K+ YouTube views and driving 40% increase in technical hiring awareness

**TechCrunch Disrupt & Startup Community** | *Mentor, Backend Engineering Track* 2018-Present

- Mentored 25+ early-stage founder teams on backend architecture, scalability strategy, and technical due diligence, with 6 mentees successfully raising Series A/B funding
- Provided architecture reviews and performance optimization consulting saving mentee companies estimated $400K+ in cloud infrastructure costs

## TECHNICAL SKILLS

**Languages** – Go, Rust, Python, Java, SQL, Bash, C++

**Technologies** – Kubernetes, Docker, gRPC, Protocol Buffers, Microservices, Event Sourcing, Distributed Systems, NoSQL (DynamoDB, Cassandra), PostgreSQL, Message Queues (Kafka, RabbitMQ), Stream Processing (Spark, Flink)

**Tools** – Git, Prometheus, Grafana, ELK Stack, Jaeger, DataDog, AWS (EC2, S3, RDS, Lambda), GCP, Terraform, Jenkins, ArgoCD, Linux kernel tuning